

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (E&E) (Credit System) Degree Examinations
Supplementary Examinations - May - June 2023

20EE502 – SIGNAL ANALYSIS AND PROCESSING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

Marks BT* CO* PO*

1. a)

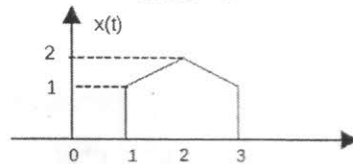


Figure 1

For the signal shown in Figure 1. Find $z(t) = 0.5x(-0.5t + 4)$.

04 L*1 1 1,2,5

b) Sketch the Even and Odd components for the following signals

i. $x(t) = t\{u(t) - u(t-1)\} + 2\{u(t-2) - u(t-4)\}$

ii. $x[n] = \cos(\pi n) \times \left(\frac{1}{4}\right)^n$ for $-2 \leq n \leq 2$

10 L2 1 1,2,5

c) Explain the process of sampling and quantization of a continuous time signal. What is the effect of under sampling?

06 L1 1 1,2,5

2. a) Determine whether or not the following signals are periodic. If periodic find the fundamental period

i) $x(t) = [\cos(2t - \frac{\pi}{3})]^2$

ii) $x[n] = 2 \cos(\frac{\pi}{4}n) + \sin(\frac{\pi}{8}n)$

08 L2 1 1,2,5

b)

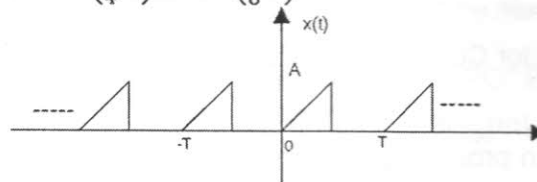


Figure 2

Figure 2 shows the periodic 50% duty cycle dc-offset sawtooth wave $x(t)$ with peak amplitude A. Determine the energy and power of $x(t)$.

08 L2 1 1,2,5

c) Consider the analog signal $x(t) = 4\cos 200\pi t + 6\sin 400\pi t + 8\cos 600\pi t$.

i) What is the Nyquist rate for this signal?

ii) Assume that we sample the signal with the sampling rate of $F_s = 500$ samples/s. What is the discrete time signal obtained after sampling?

04 L1 1 1,2,5

Unit – II

3. a) Determine whether the following system is

i. Memoryless ii. Time invariant iii. Linear and Justify your answers.

$$y(t) = \int_{-\infty}^{2t} x(\tau) d\tau$$

06 L3 2 1,2,5

b)

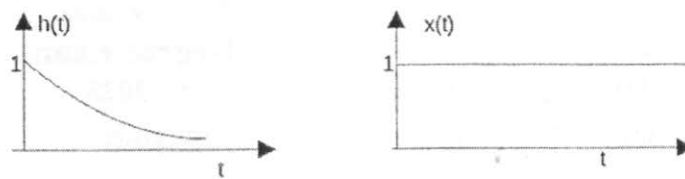


Figure-3

Compute the response of an LTI system described by its impulse response $h(t) = e^{-at}u(t)$, $a > 0$ to the step input $x(t) = u(t)$ as show in Figure 3.

06 2 2 1,2,5

c)

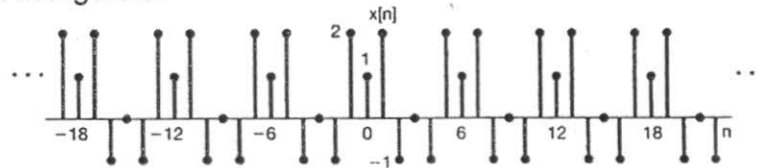


Figure 4

Compute Fourier series for the signal $x[n]$ shown in Figure 4

08 3 2 1,2,5

4. a) Determine the Fourier series representation of the signal $x(t)$ shown in Figure 5.

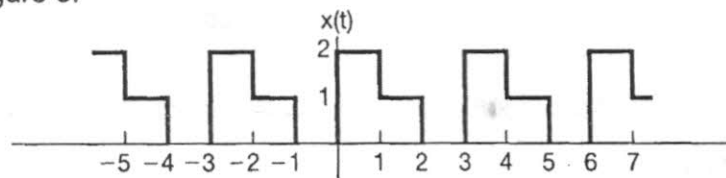


Figure 5

08 3 3 1,2,5

- b) Compute the convolution sum using $x[n] = (0.8)^n u[n]$ and $h[n] = (0.3)^n u[n]$

08 1 3 1,2,5

- c) Explain the Dirichlet conditions for Convergence of a Fourier Series.

04 3 3 1,2,5

Unit - III

5. a) State and prove the convolution property for Continuous time Fourier Transform.

06 1 3 1,2,5

- b) Find Fourier transform of the signal $x(t) = \{te^{-2t}\sin 4t\}u(t)$.

08 2 3 1,2,5

- c) Find the impulse response of the causal LTI system described by the following differential equation.

$$\frac{d^2 y(t)}{dt^2} + 6 \frac{dy(t)}{dt} + 8y(t) = 2x(t)$$

06 3 3 1,2,5

6. a) State and Prove time shift property of DTFT.

05 1 3 1,2,5

- b) Compute Fourier Transform of the following signal.

$$x[n] = \left(\frac{1}{3}\right)^{|n|} u[-n-2]$$

07 2 3 1,2,5

- c) Consider discrete LTI system with impulse response.

$$h[n] = \left(\frac{1}{2}\right)^n u[n]$$

Using Fourier Transform find the response to $x[n] = \left(\frac{3}{4}\right)^n u[n]$.

08 2 3 1,2,5

Unit – IV

7. a) Compute 4 point circular convolution of the following sequences using time domain approach.
 $x_1[n] = \{1, 2, 3, 1\}$ and $x_2[n] = \{4, 3, 2, 2\}$
- b) Compute 8 point DFT of a real sequence $x[n] = 1$ for $0 \leq n \leq 7=0$ otherwise using decimation in frequency FFT algorithm.
- c) Calculate the percentage saving in additions in a 512 point radix-2 FFT when compared to direct DFT.
8. a) Find 8 point DFT of a real sequence $x[n] = [0.5 \ 0.5 \ 0.5 \ 0.5 \ 0 \ 0 \ 0 \ 0]$ using decimation in time FFT algorithm.
- b) Determine the eight point DFT of $x[n] = [1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 0]$. Sketch the magnitude and phase spectrum.

06	3	4	1,2,5
10	1	4	1,2,5
04	3	4	1,2,5
10	2	4	1,2,5
10	2	4	1,2,5

Unit – V

9. a) Explain the frequency mapping of S plane into Z plane in a bilinear transformation.
- b) Design a linear phase FIR low pass filter using rectangular window by taking 7 samples of window sequence and with cutoff frequency, $\omega_c = 0.2\pi$ rad/sample
10. a) Design a Butterworth digital low pass filter using bilinear transformation by taking $T = 0.1$ seconds to satisfy the following specification.
- b) Draw Direct I structure of digital IIR low pass filter.

$$0.6 \leq |H(e^{j\omega})| \leq 1.0; \text{ for } 0.7\pi \leq \omega \leq \pi$$

$$|H(e^{j\omega})| \leq 0.1; 0 \leq \omega \leq 0.35\pi$$

$$H(z) = \frac{0.5073z^{-1} + 0.1985z^{-2}}{1 - 0.536z^{-1} + 0.3055z^{-2} - 0.0551z^{-3}}$$

08	3	5	1,2,3
12	3	5	1,2,3
14	3	5	1,2,3
06	2	5	1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
